Cosine rule

1 $\cos\theta = \dfrac{b^2 + c^2 - a^2}{2bc}$

2

a $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

b $a^2 = b^2 + c^2 - 2bc\cos A$

c $b = \sqrt{a^2 + c^2 - 2ac\cos B}$

3

a $\cos Q = \dfrac{p^2 + r^2 - q^2}{2pr}$

b $r^2 = p^2 + q^2 - 2pq\cos R$

4 $\angle ABC$

Unknown sides

5

a $a = 14.14$

b $a = 212.34$

c $q = 19.23$

d $c = 10.60$

e $a = 14.11$

f $c = 13.17$

g $y = 6.52$

h $y = 49.40$

i $z = 21.67$

j $w = 12.75$

6 $AB = 3\sqrt{2}$

7 $v = 5$

Unknown angles

8

a $w = 42^\circ$

b $w = 32^\circ$

c $A = 27^\circ$

d $A = 34^\circ$

e $A = 39^\circ$

f $B = 51^\circ$

g $B = 53^\circ$

h $C = 104^\circ$

9 $\theta = 48.24^\circ$

10 $B = 41^\circ 47' 4''$

11 $\cos Q = -\frac{5}{}$

12 $\theta = 24^{\circ}$

13 $x = 142^{\circ}$

Applications

14 $x = 34.78 \text{ km}$

15 $x = 40^{\circ}$

16 $x = 48.57$

17 $x = 45.7 \text{ cm}$

18

a $x = 277.3 \text{ km}$

b 568.2 km

19 $x = 53.6 \text{ m}$